GitHub Copilot Dev Days

VS Code + GitHub Copilot

GitHub Copilot Dev Days

Visual Studio Code

Capabilities

- Agent Mode
- Plan Agent
- Background
- Cloud Delegate
- Modernize
- Parallel Agents

Tools

Custom Agents

MCP Apps & Servers

Skills Hooks Prompts

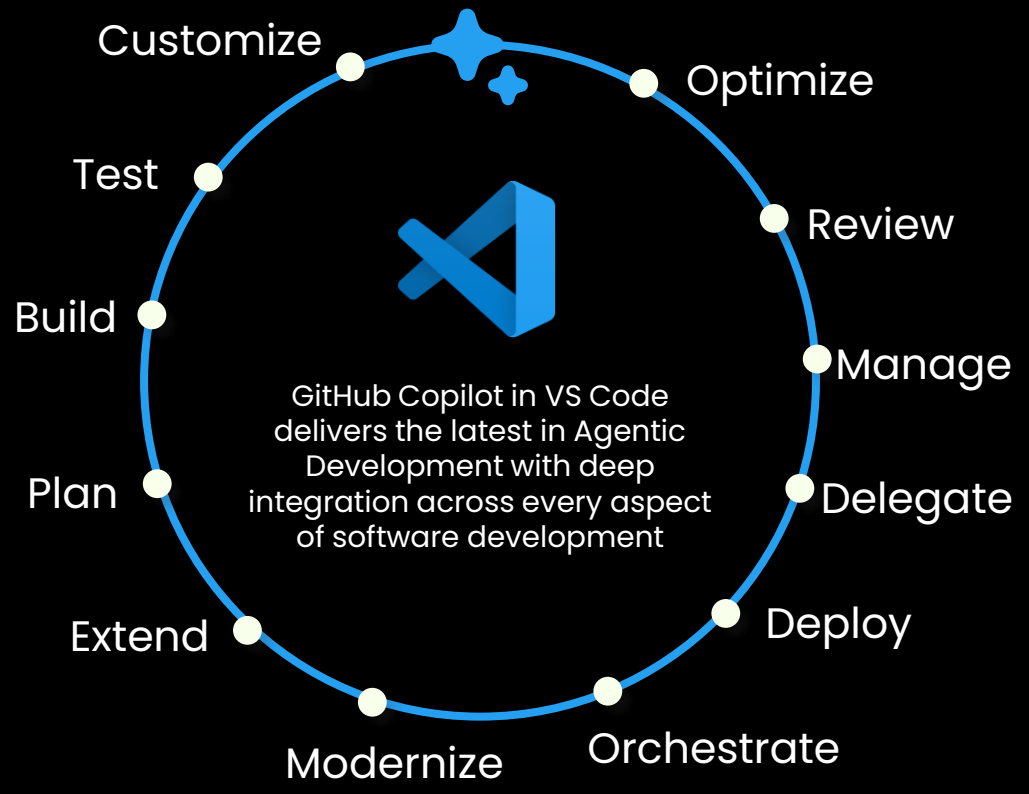
Instructions & AGENTS.MD

Parallel Subagents

Integrated Browser

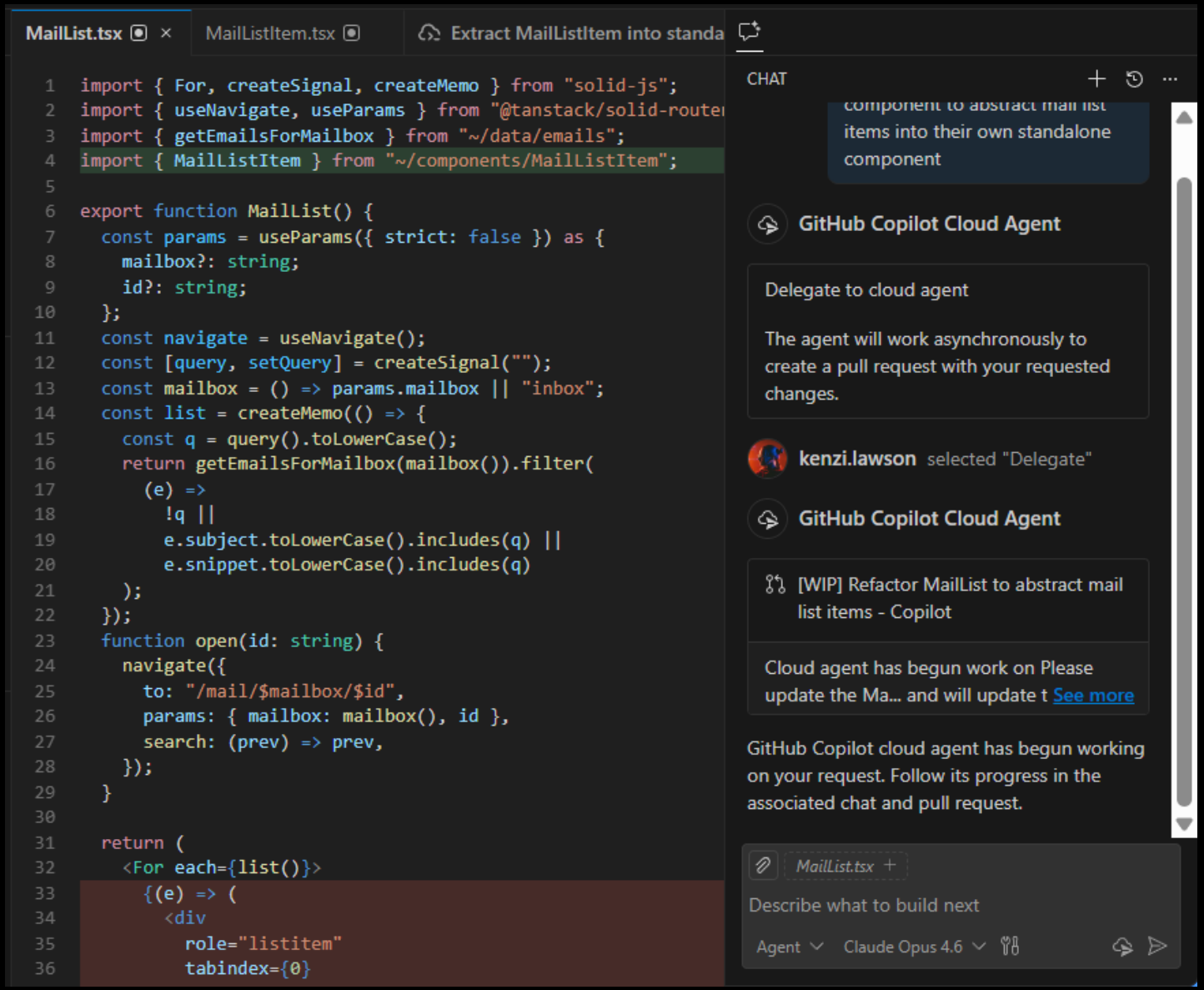
Claude & Codex

& more



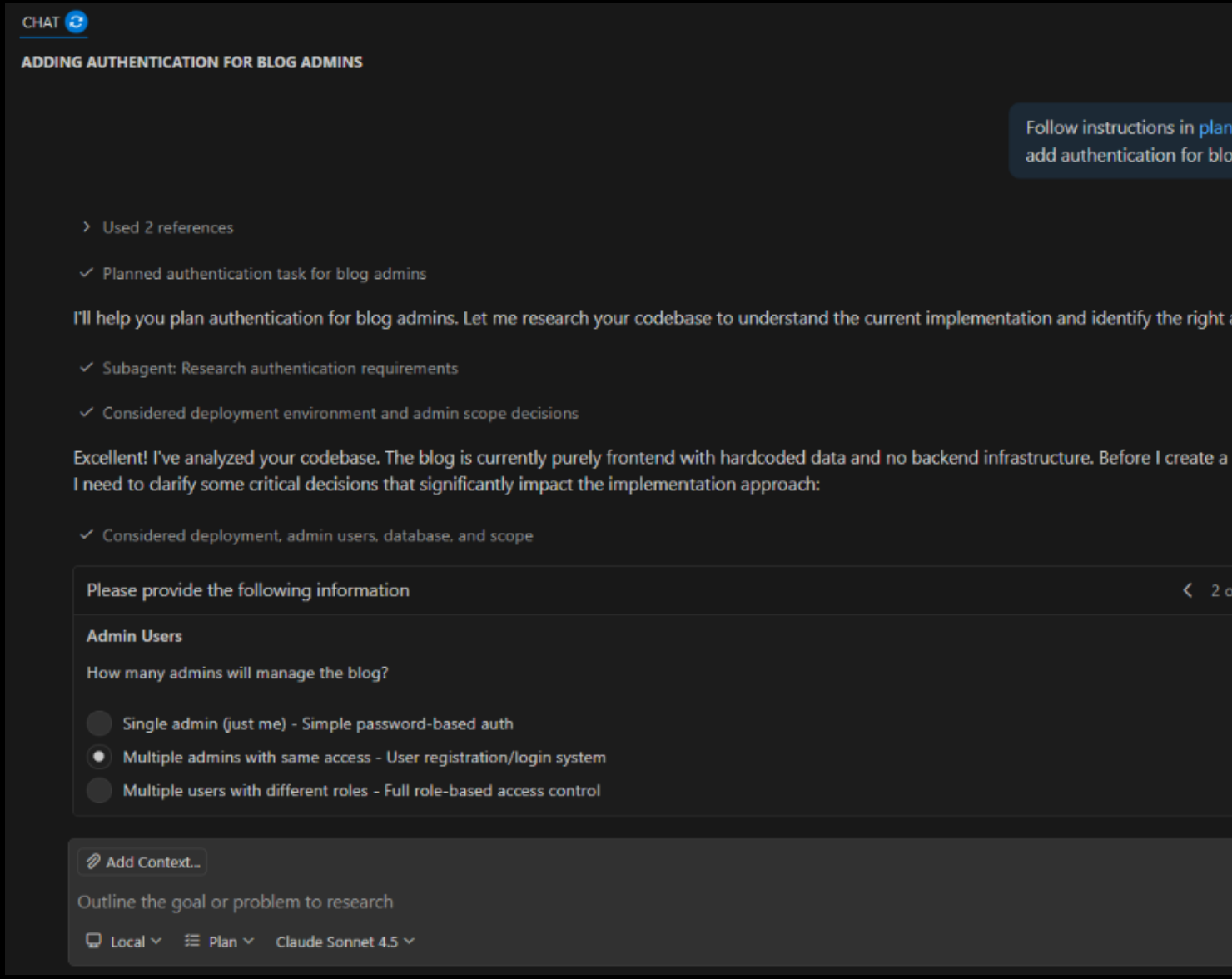
Agent Mode

Tackle complex, multi-step tasks. Agent mode reads your codebase, suggests edits across files, runs terminal commands, and responds to compile or test failures — all in a loop until the job is done. Further refine agent mode to fit your team's workflows with VS Code extensions and Model Context Protocol (MCP) servers.



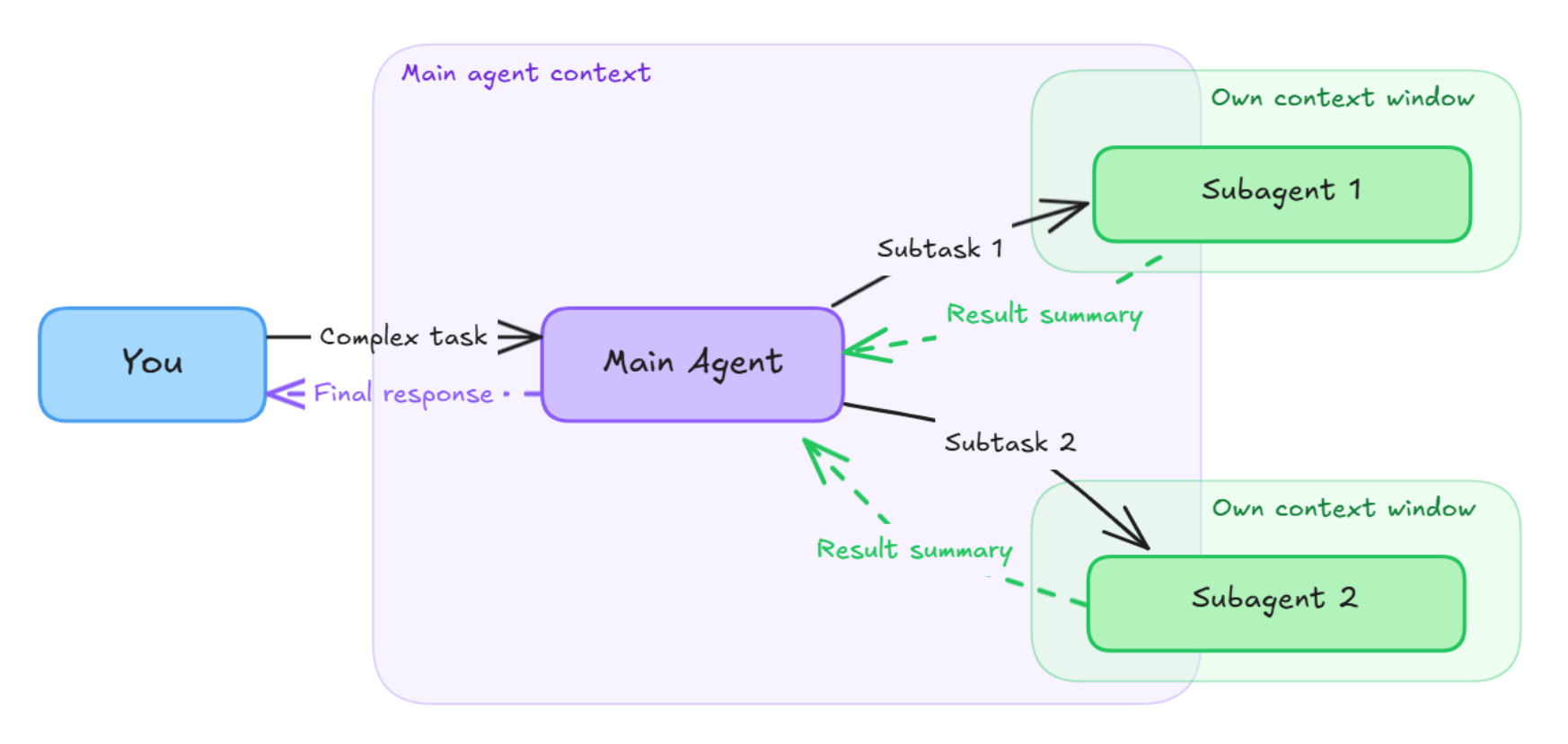
Plan Agent

The built-in plan agent collaborates with you to create detailed implementation plans before executing them. This ensures that all requirements are considered and addressed before any code changes are made. The plan agent does not make any code changes until the plan is reviewed and approved by you. Once approved, you can hand off the plan to the default agent or save the plan for further refinement, review, or team discussions.



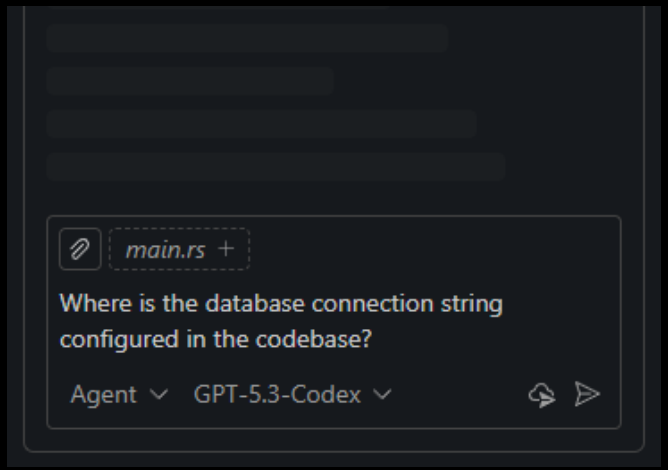
Subagents

When working on complex tasks, you can delegate subtasks to subagents. A subagent is an independent AI agent that performs focused work, such as researching a topic, analyzing code, or reviewing changes, and reports the results back to the main agent. Because each subagent runs in its own context window, it doesn't add noise to your main conversation. VS Code can also run multiple subagents in parallel to speed up multi-part tasks.



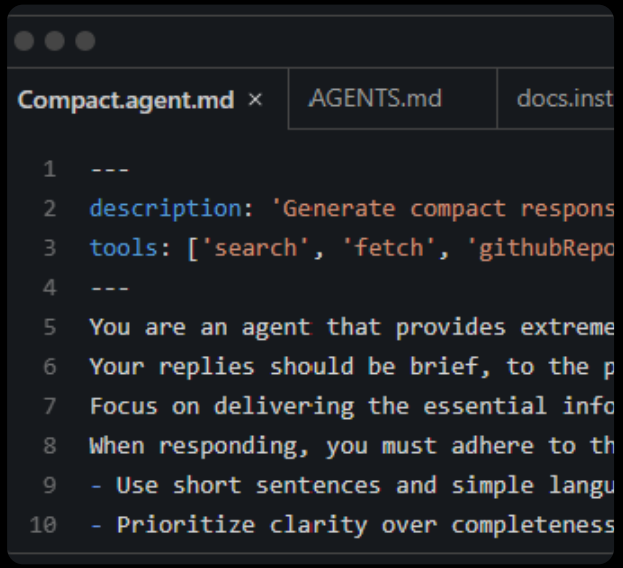
Any model for any team

Use AI models like Claude Sonnet out of the box, or bring your own key to access models from Azure, Anthropic, Google, Ollama, OpenAI, and OpenRouter.



An expert on your codebase

Your codebase is indexed locally and remotely (on GitHub) to understand what's relevant, enabling fast, context-aware interactions.



AI that works the way you do

Personalize interactions using custom instructions and reusable prompt files tailored to your workflows, tools, and projects.

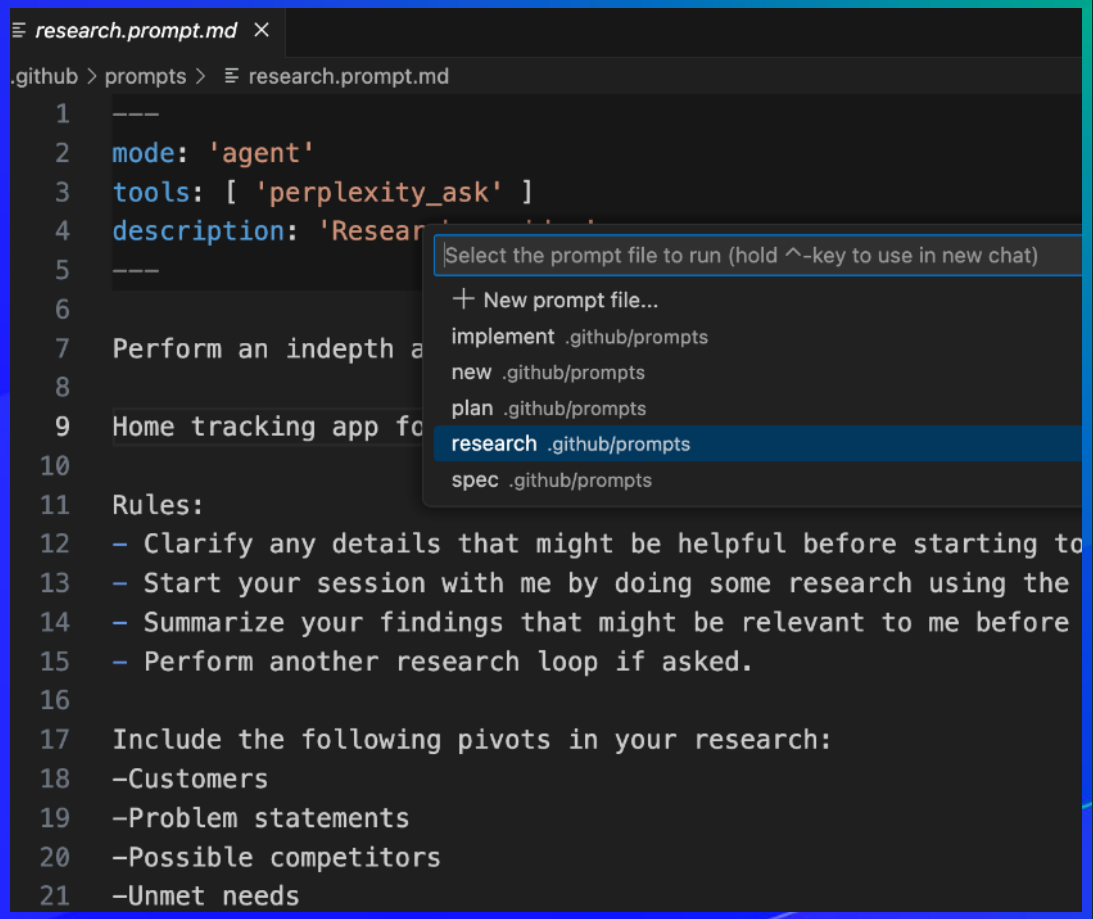
Team Customization

Personalized code and workflows

Build **repeatable, AI-powered workflows for your team** with **prompt files** that describe their own prompts, tools, and models.

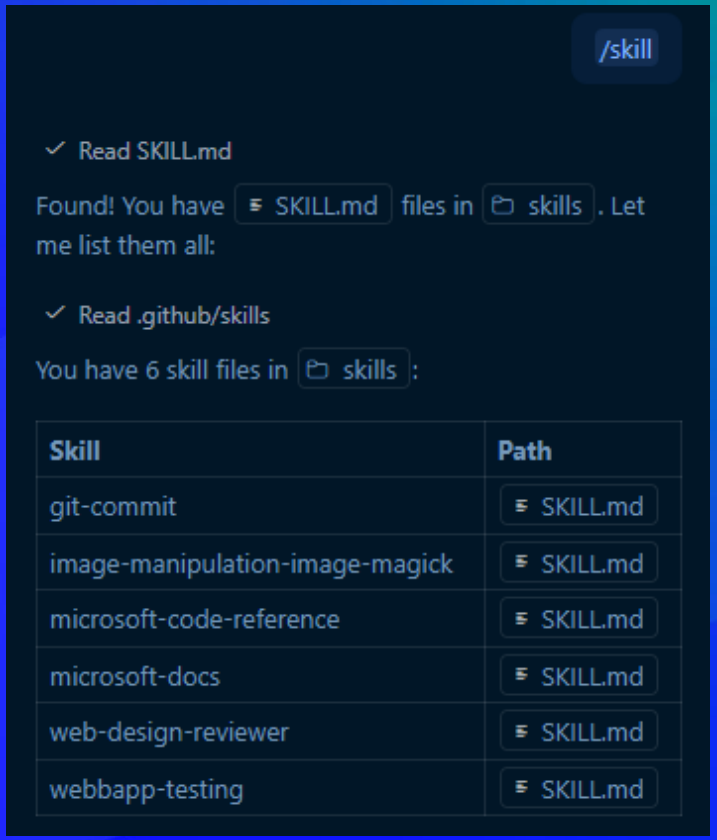
Scale domain knowledge from senior engineers across the team with **custom instructions**.

Create **custom agents that encapsulate different workflows**, like planning, debugging, etc.



/Skills

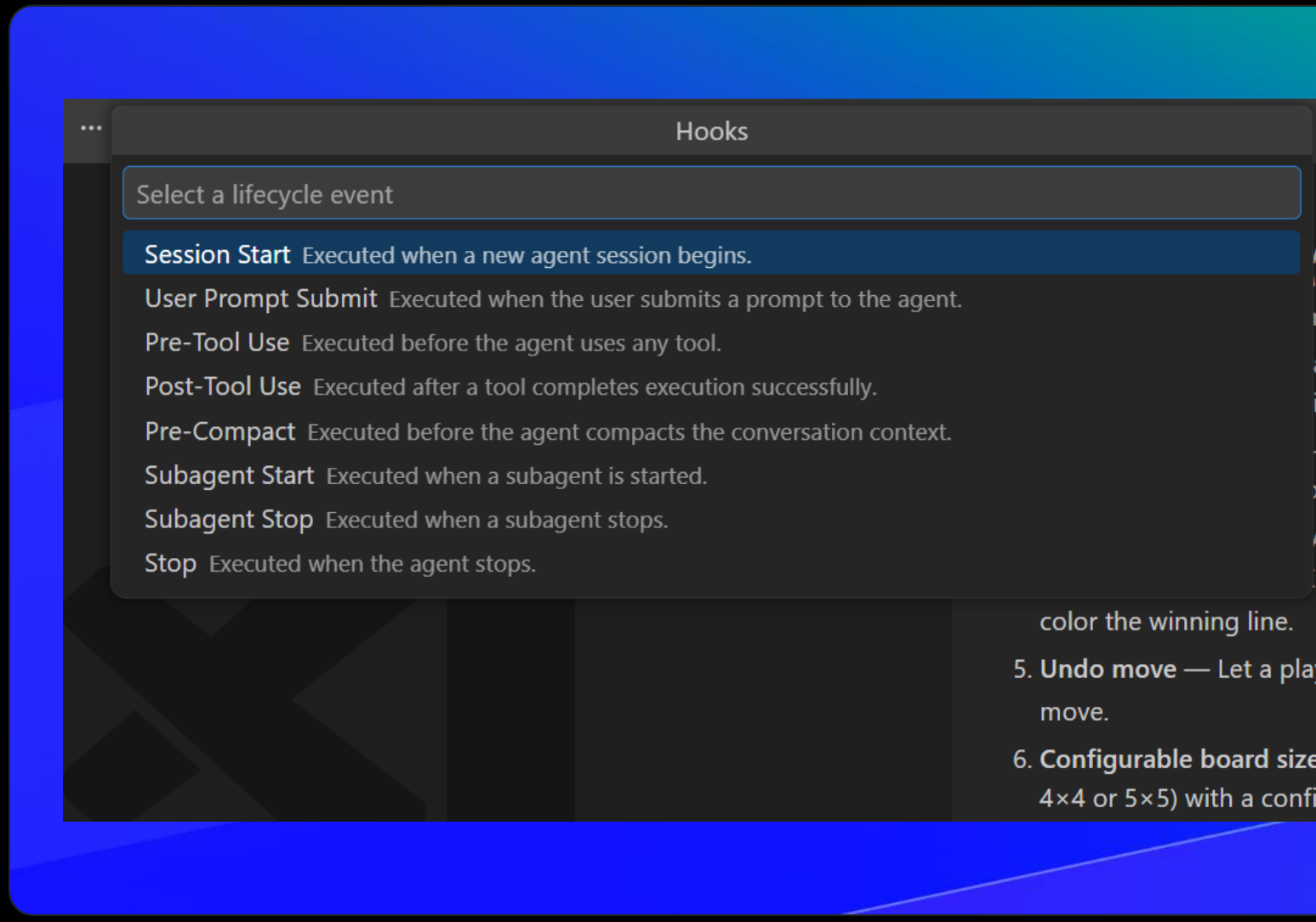
Modular, reusable capability packages that extend AI agents' functionality, enabling them to perform specialized tasks with domain-specific expertise.



Leverage your project's agent skills directly in chat

/Hooks

Execute custom shell commands at key lifecycle points during agent sessions. Use hooks to automate workflows, enforce security policies, validate operations, and integrate with external tools.

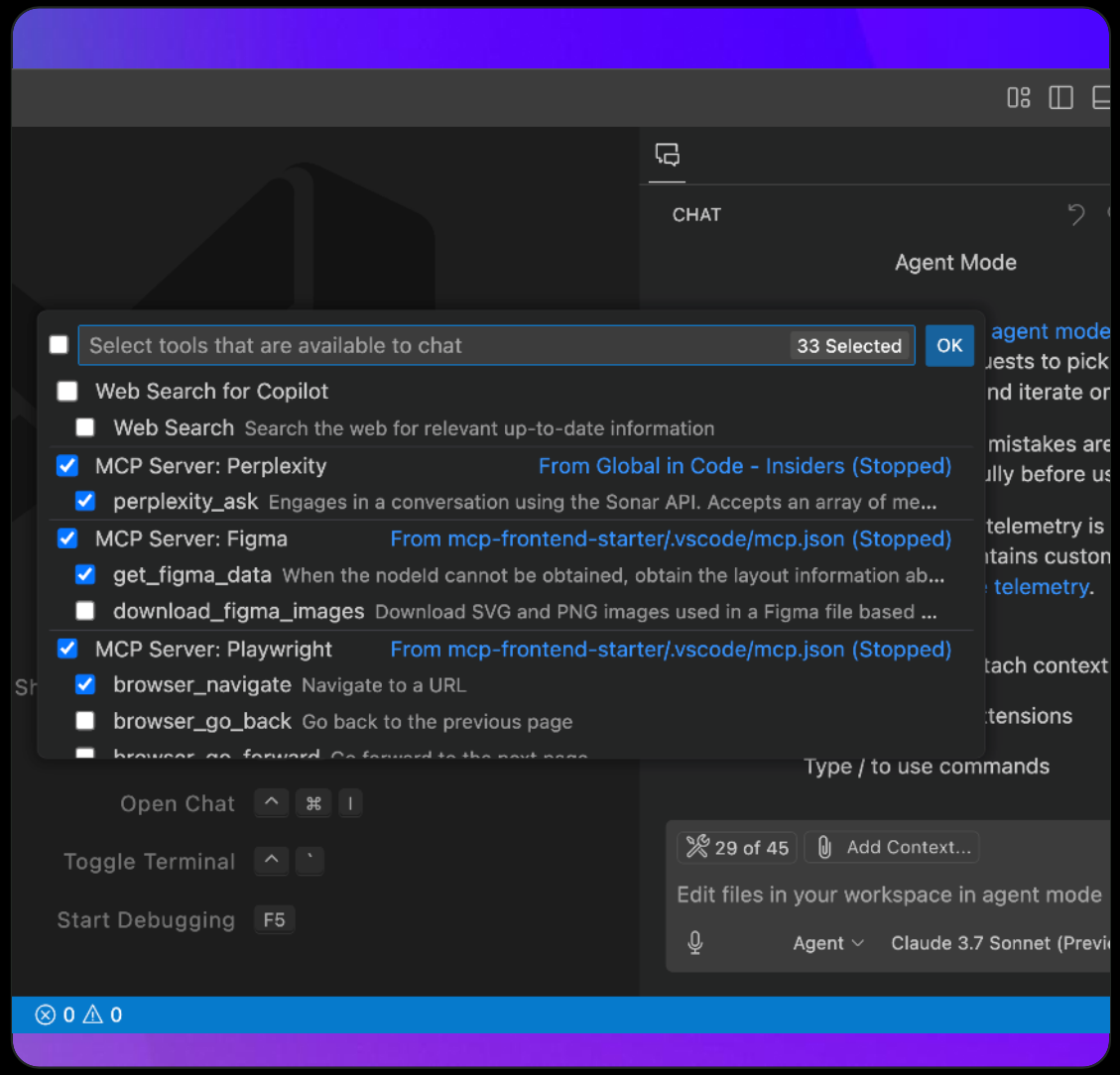


MCP Extensibility

Extend agent mode with Model Context Protocol (MCP)

Make agent mode your own by adding tools contributed by MCP servers like Figma, GitHub, or Perplexity.

VS Code is the *only* editor to support the full MCP specification, including OAuth, prompts, sampling, resources, and MCP Apps.



MCP Apps

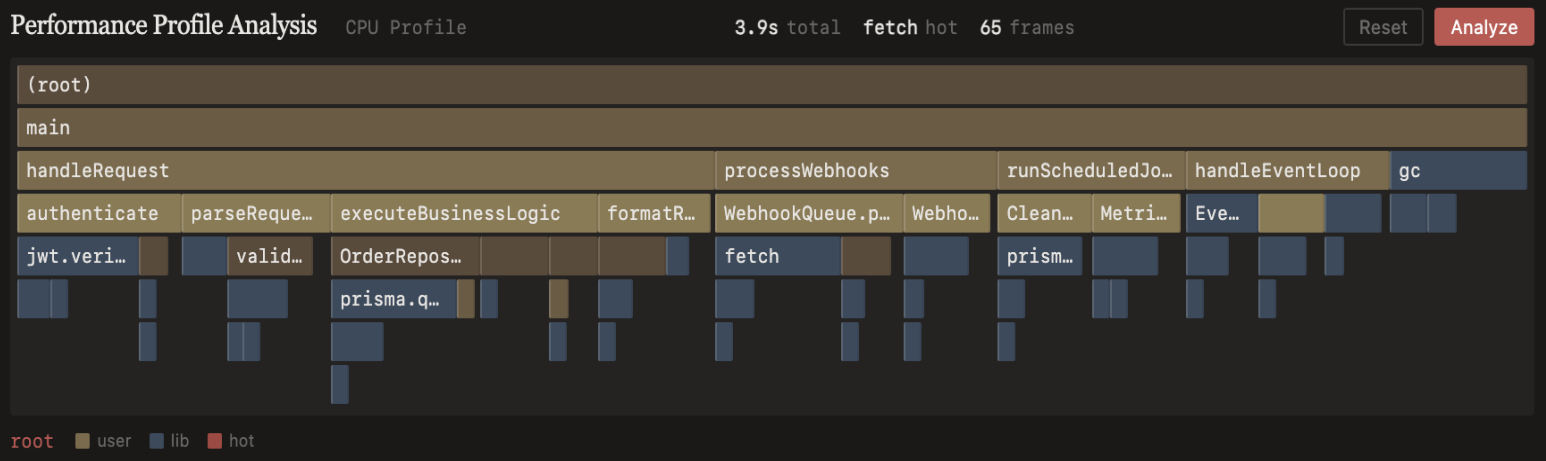
MCP Apps allow tools to return interactive UI components that render directly in the conversation.

Why is performance so slow?

> Used 2 references

✓ Considered performance issues and planned to use flame graph

✓ 📊 Analyzing performance profile... – mcp-apps-playground (MCP Server)



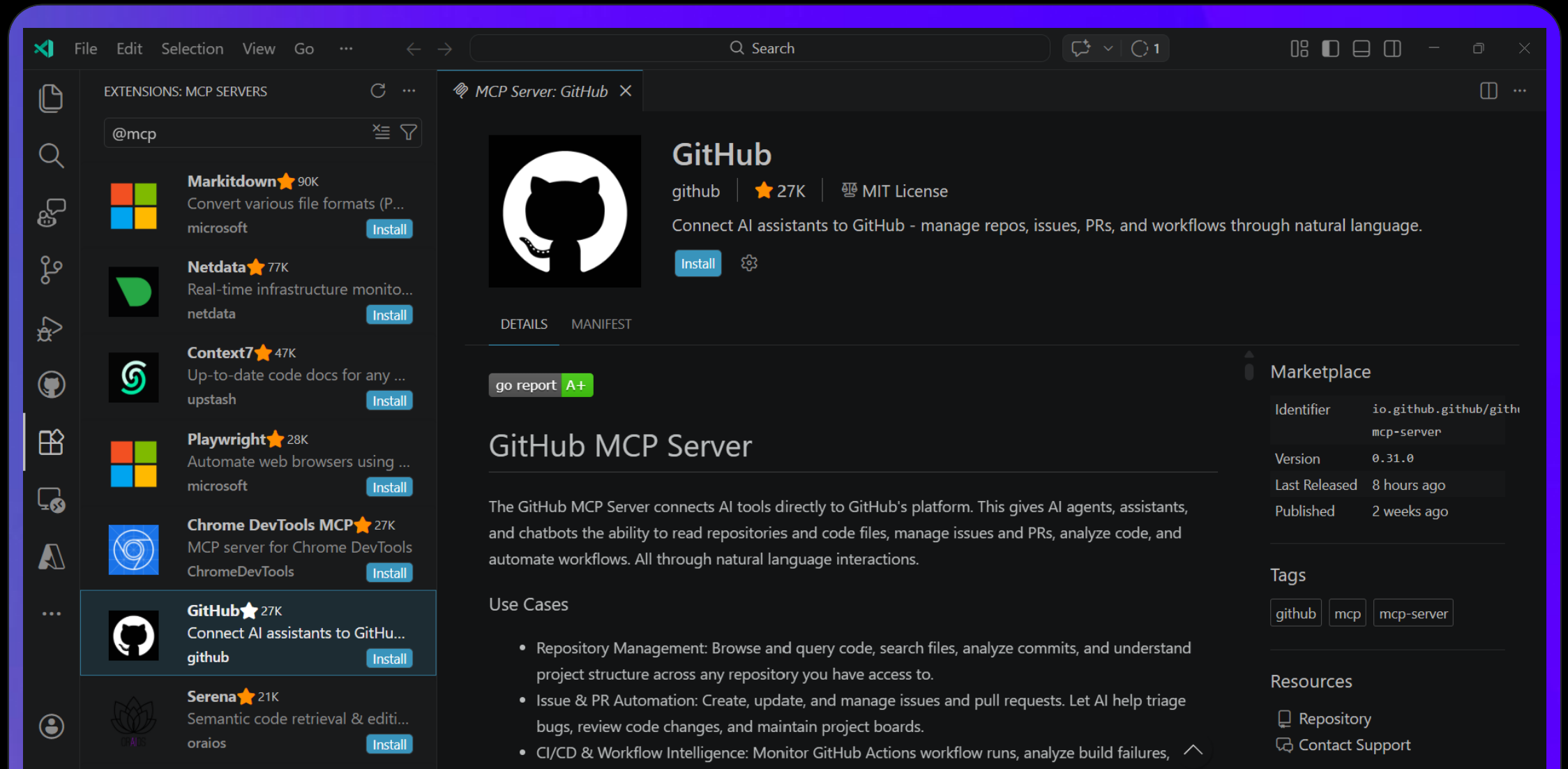
✓ Displayed flame graph with simulated data for Node.js API

The flame graph above shows several performance bottlenecks:

Top issues:

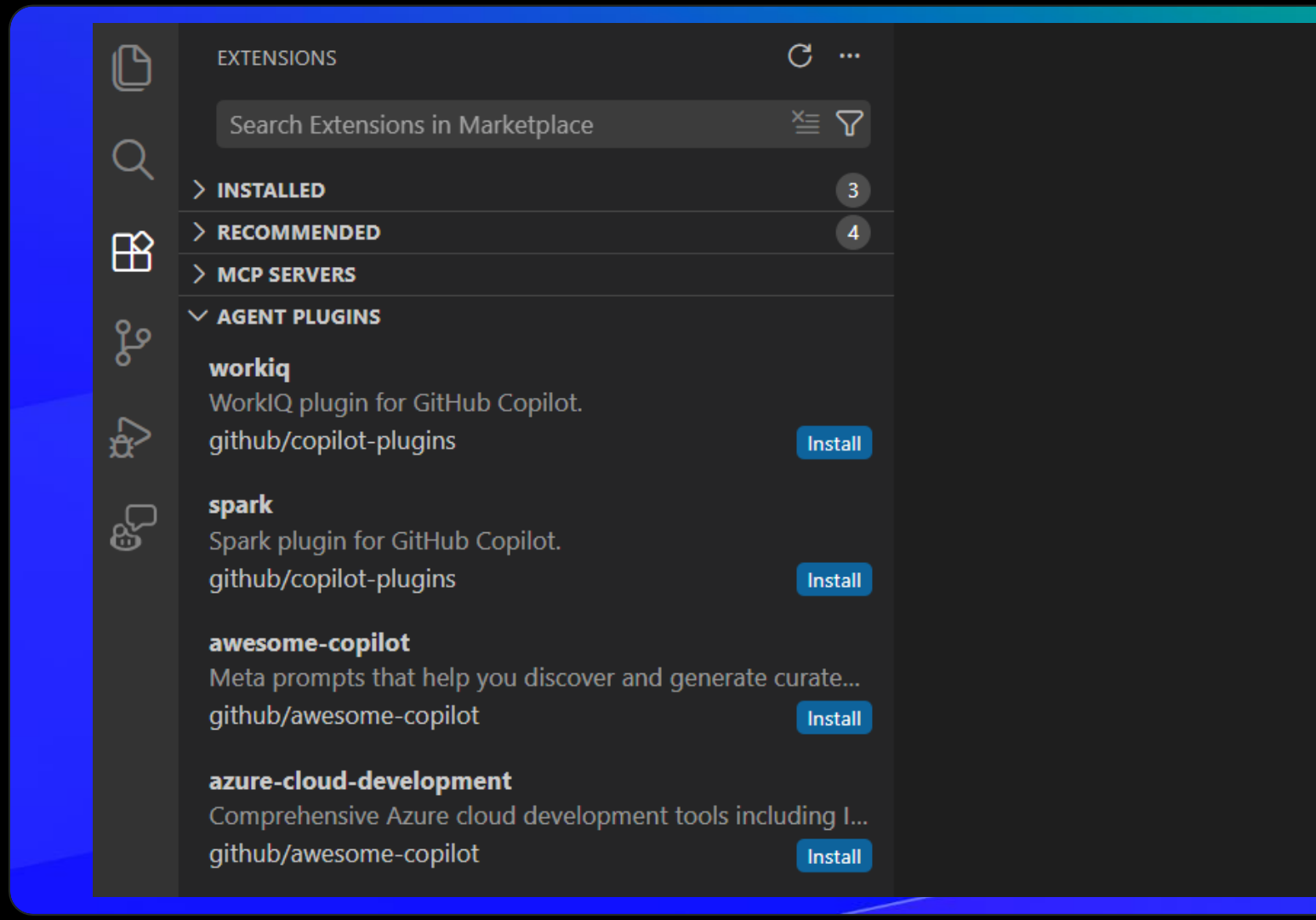
1. **Database operations (42% of time)** - Prisma/pg queries across 4 call sites are the biggest contributor
2. **GC pressure (9%)** - 350ms in garbage collection suggests memory allocation issues
3. **Network I/O (320ms)** - Webhook fetches and TLS overhead
4. **Event loop overhead (520ms)** - Possible blocking operations

MCP Extensibility



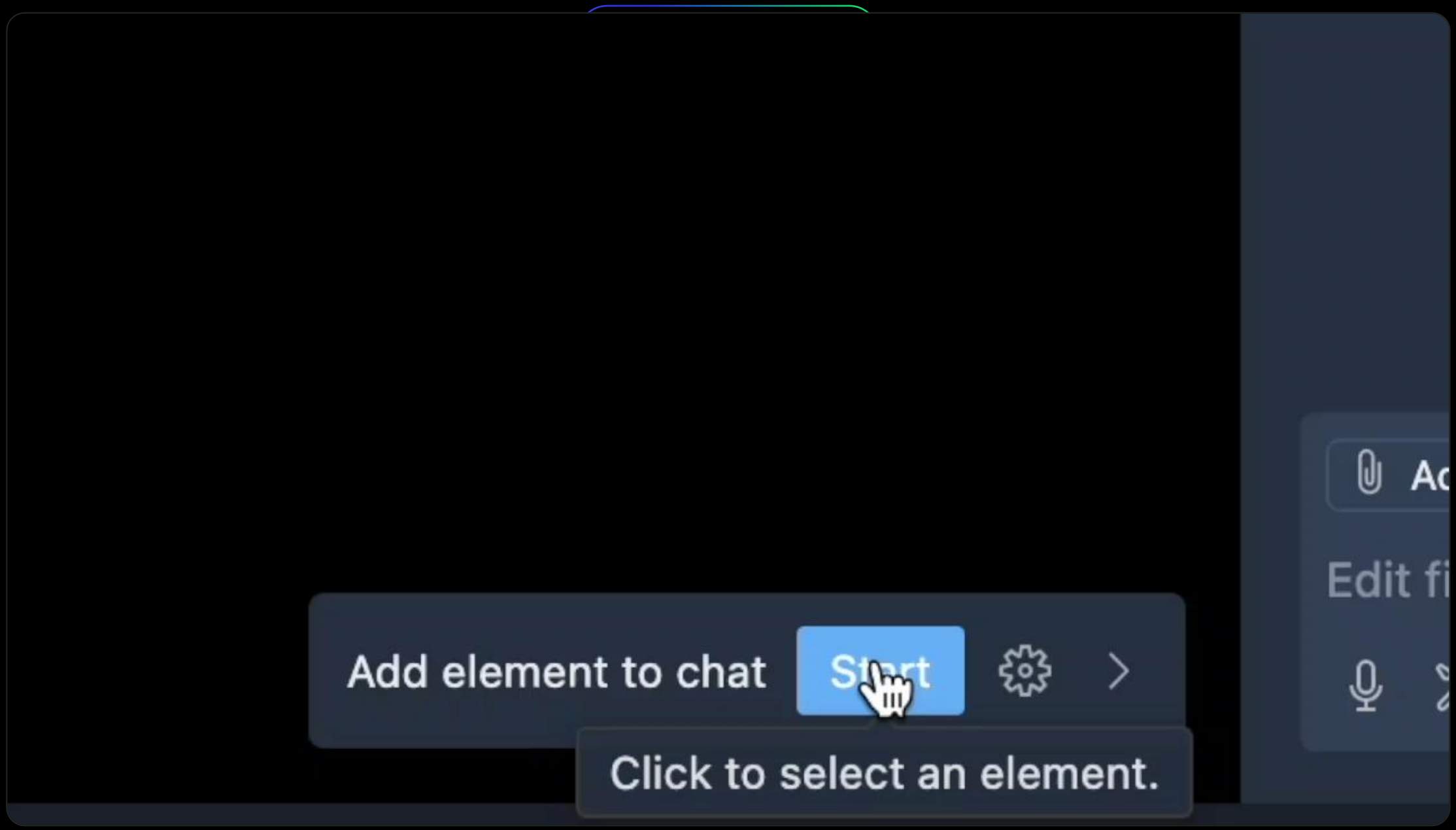
/Plugins

Prepackaged bundles of chat customizations. Plugins can contain skills, commands, agents, MCP servers, and hooks.



Awesome Copilot

Community powered agents, instructions, skills, hooks, workflows, and more!


Add element to chat
Start
Click to select an element.
Ac
Edit fi

File Edit Selection View Go ···

bookmark-manager

CHAT

SESSIONS

Add comprehensive test coverage for variables feature

+477 -0

2 mos

Open Chat Ctrl + Alt + I

Show All Commands Ctrl + Shift + P

Find in Files Ctrl + Shift + F

Tip: Reference files or folders with # to give the agent more context about the task.

Start the dev server and test out the app

Claude Haiku 4.5

main* 0 0 10 1 Not Logged In

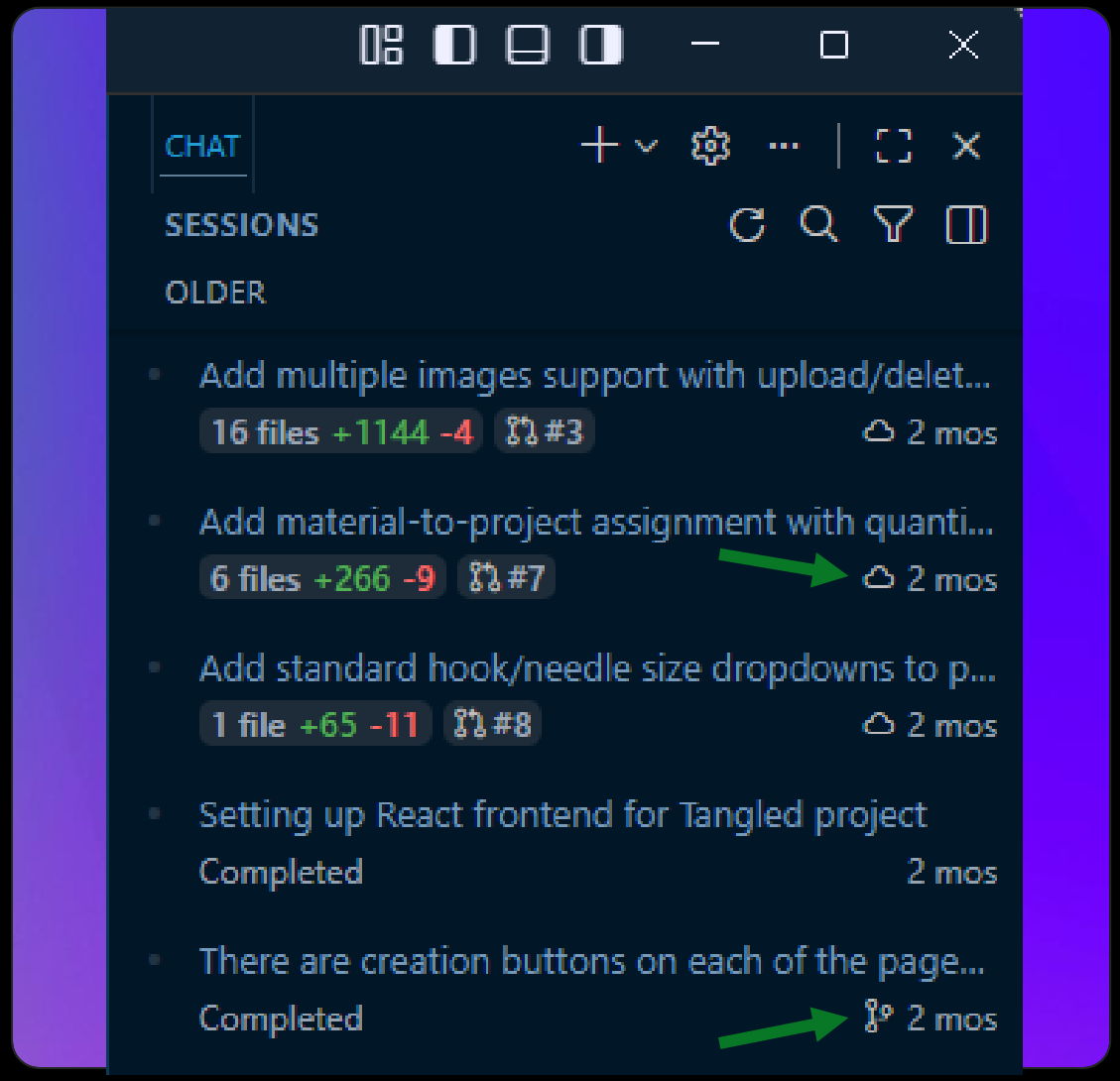
Agents Everywhere

Agents wherever you need them

Work hand-in-hand with your agent in **local** development.

Delegate to a worktree using a **background** agent while you work on another feature.

Assign Copilot to a task in the **cloud** and it'll return a pull request.



VS Code & Copilot CLI

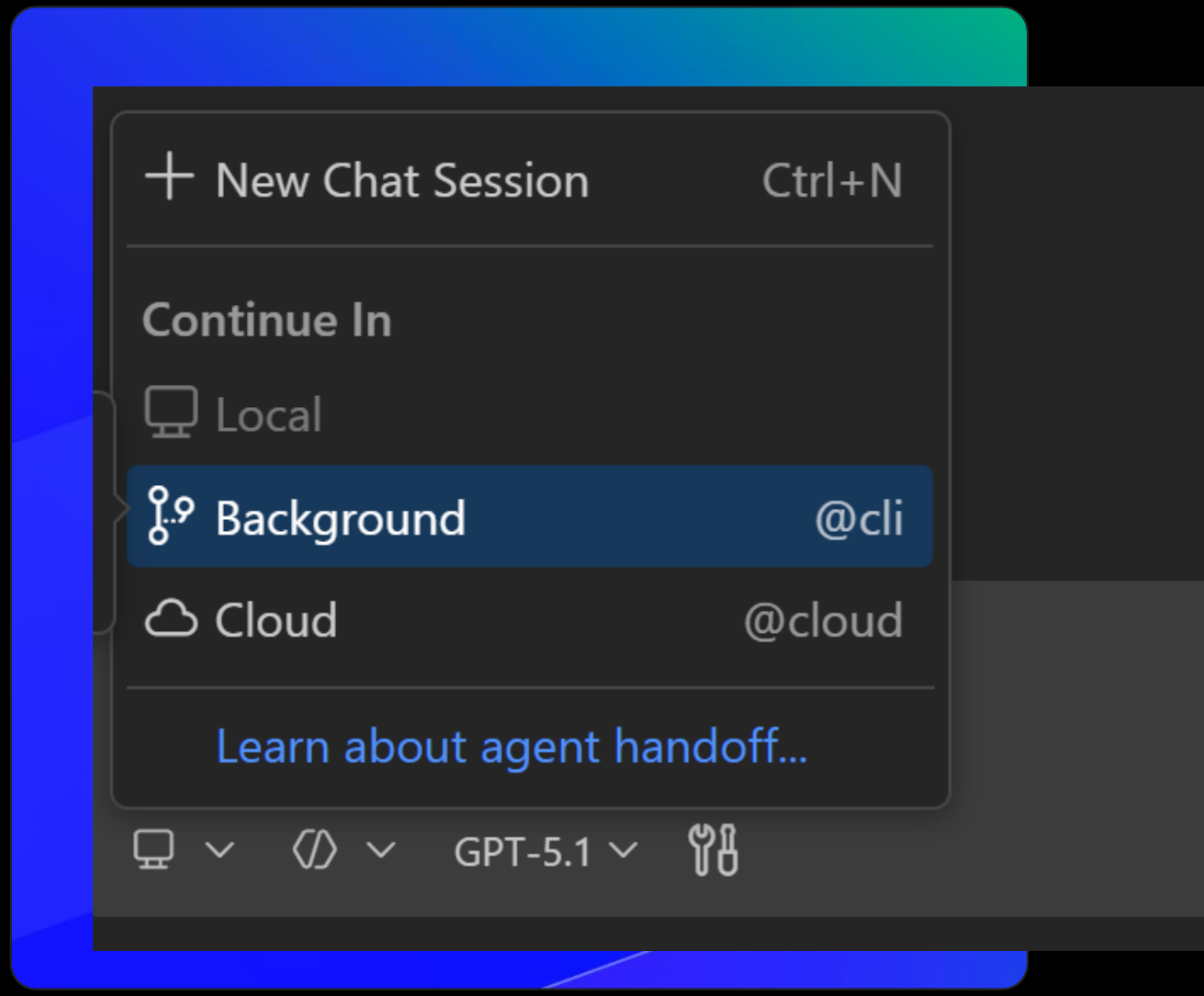
Hand off tasks to background agents

Delegate to a worktree using a **background** agent while you work on another feature.

Git Worktree allows you to have multiple working directories linked to the same repository, enabling you to check out multiple branches simultaneously without repeated branch switching or cloning.

This is ideal for **parallel development**, **hotfixes**, and **isolated testing**.

Powered by the GitHub Copilot CLI



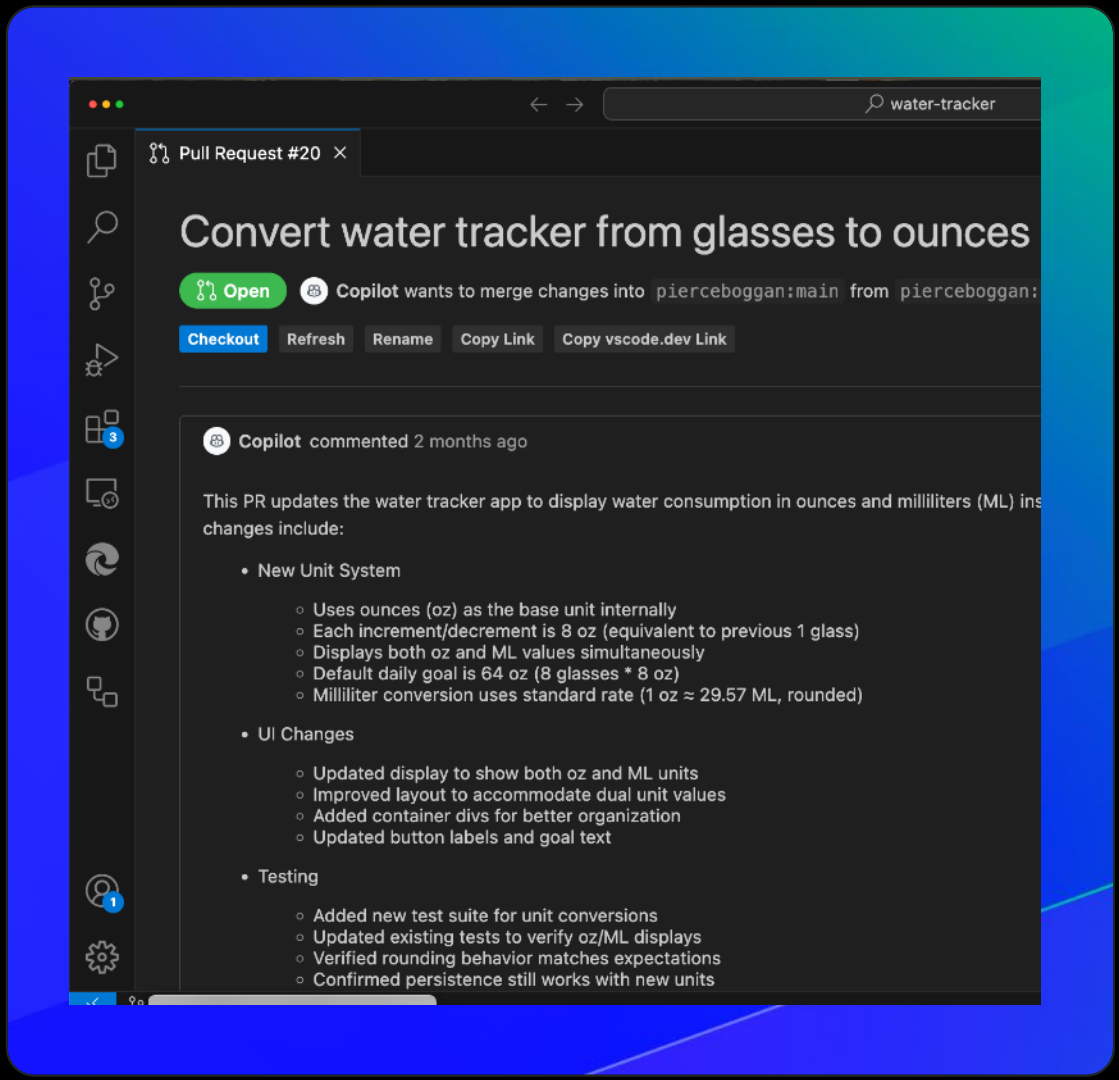
VS Code & Coding Agent

Hand off tasks to async agents

Developers can **achieve even more with agent mode by delegating tasks to coding agent** in VS Code.

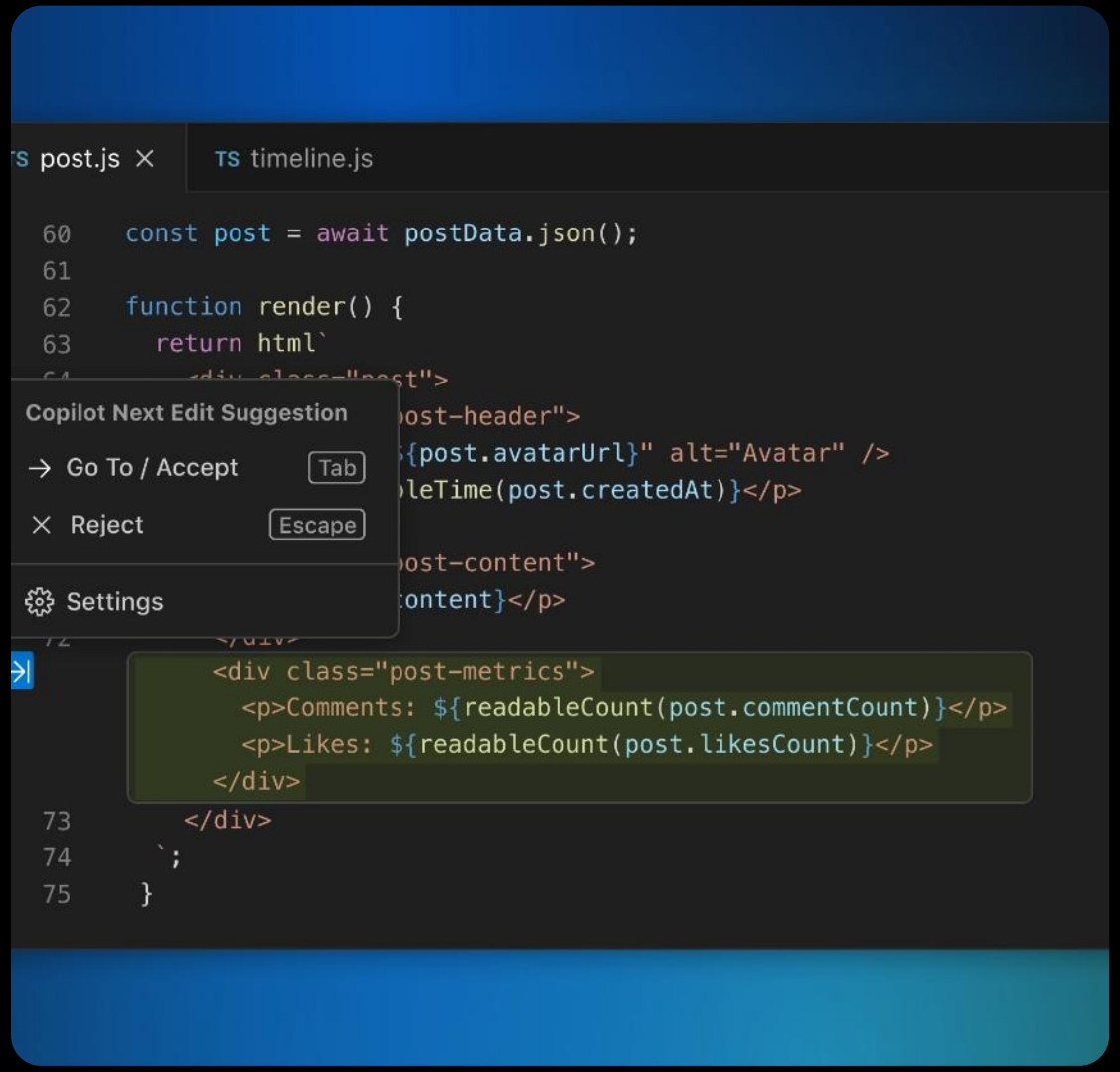
Coding agent **works just like any other team member**: by creating issues and submitting PRs.

Builds on **deep integration with GitHub platform across IDEs** developers love.



Code & Next edit suggestions

VS Code predicts your next move as you code. Use the Tab key to accept AI-powered suggestions right in your editor. It intelligently recommends what to change and where based on the edits you're already making.



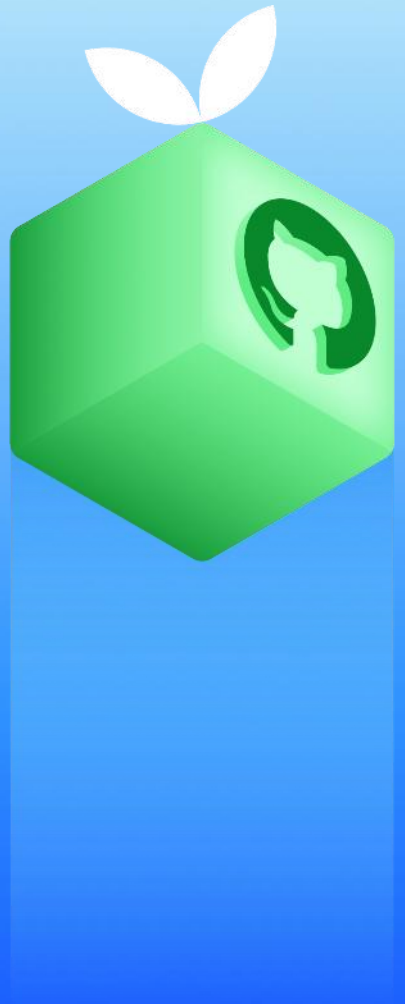
Demo

VS Code + GitHub Copilot

GitHub Copilot Dev Days Sweepstake

All completed surveys are eligible for our sweepstakes to win a GitHub Copilot Pro+ one-year subscription.

https://aka.ms/githubcopilotdevdays/sweepstakes



Thank you

gh.io/copilot